FOPRA 08: High Resolution X-Ray Diffraction

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March 17, 2026

Question 1: *Plot the long $2\theta/\omega$ scan of sample B in a logarithmic scale. Which peaks can you identify?*

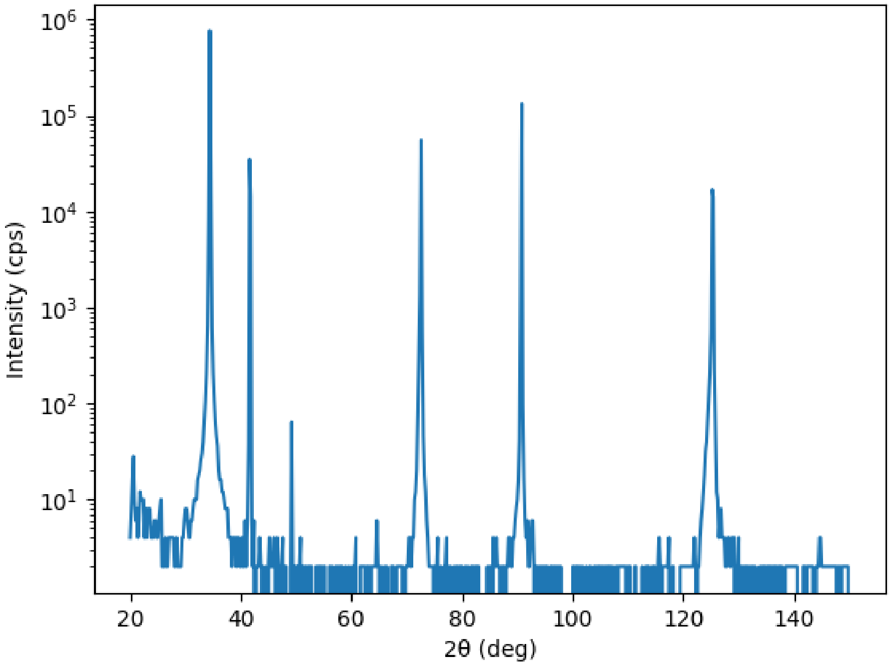

Figure 1: $2\theta/\omega$ wide range scan of sample B (002)

The peaks correspond to Bragg reflections of the ZnO film and the Al_2O_3 substrate.

- $2\theta = 34.4°$ → ZnO (002)
- $2\theta = 41.6°$ → Al_2O_3 (006)
- $2\theta = 72.6°$ → ZnO (004)
- $2\theta = 90.8°$ → Al_2O_3 (0012)
- $2\theta = 125.2°$ → ZnO (006)

Only (00l) reflection appear, indicating that both ZnO film and Al_2O_3 substrate are c-axis oriented.

Question 2: *Estimate the layer thickness of sample A by using Equation (41).*

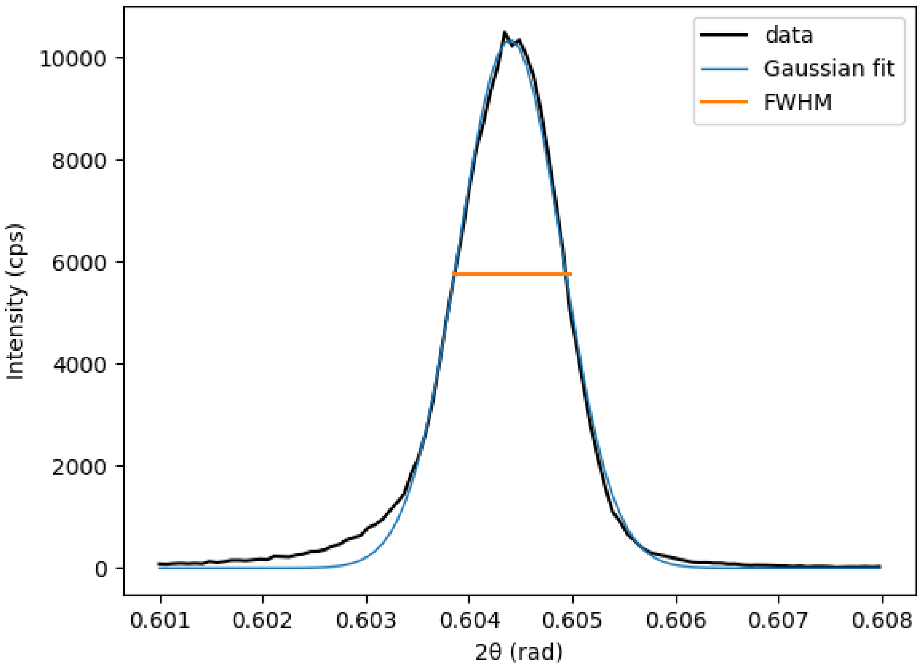

Figure 2: $2\theta/\omega$ scan of sample A (002) and Gaussian fit

According to equation (41):

$$l_z = \frac{0.9\lambda}{\Delta(2\theta)\cos\theta} \tag{1}$$

and $\Delta(2\theta) \approx 0.0011$ rad from Gaussian fit, layer thickness l_z is estimated as 130.023 nm.

Question 3: *Determine the layer thickness of sample B by using Equations (41) and (42). Compare the obtained values! Conclusion?*

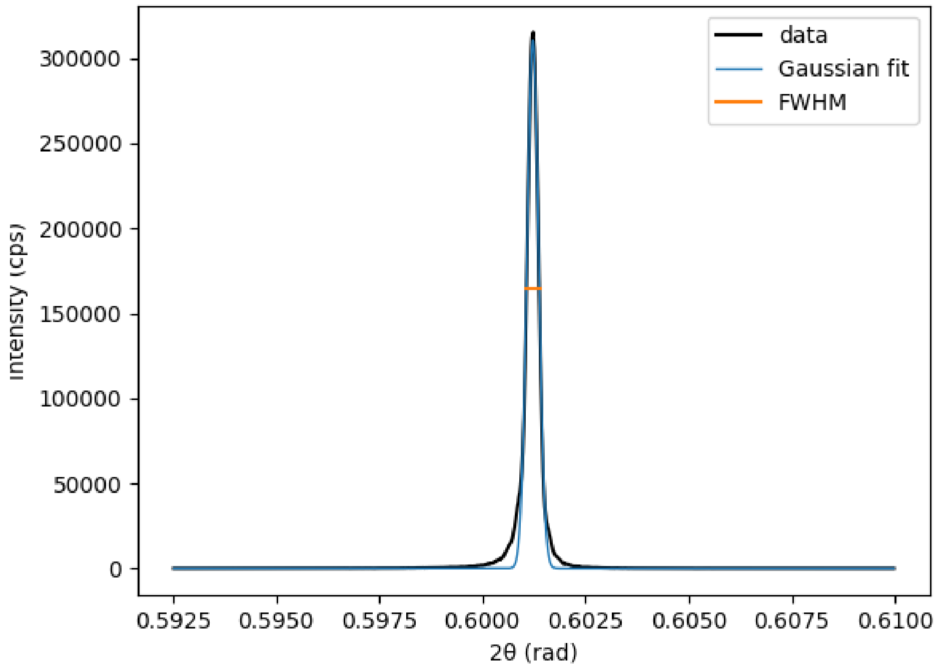

Figure 3: $2\theta/\omega$ scan of sample B (002) and Gaussian fit

According to equation (41) (equation (1) in this report) and $\Delta(2\theta) \approx 0.0003$ rad from Gaussian fit, layer thickness l_z is estimated as 452.025 nm.

Equation (42):

$$l_z = \frac{\lambda \sin\theta}{\Delta(\theta)\sin 2\theta} \tag{2}$$

requires the distance between secondary maxima. However, in the measured $2\theta/\omega$ scan of sample B, only a single peak is visible and no clear secondary maxima can be identified. Therefore, equation (42) cannot be applied to this dataset.

This suggests that the thickness estimated from equation (41) is the only accessible one for sample B from this measurement.

Question 4: *Estimate the screw-type dislocation densities of samples A and B and their lateral crystallite size. Compare the structural quality of both samples.*

By assuming that the broadening of rocking curve is only caused by limited size, lower limit for the lateral crystallite size $l_{\|}$ can be estimated by:

$$l_{\|} = \frac{0.9\lambda}{\Delta\omega \sin\theta} \tag{3}$$

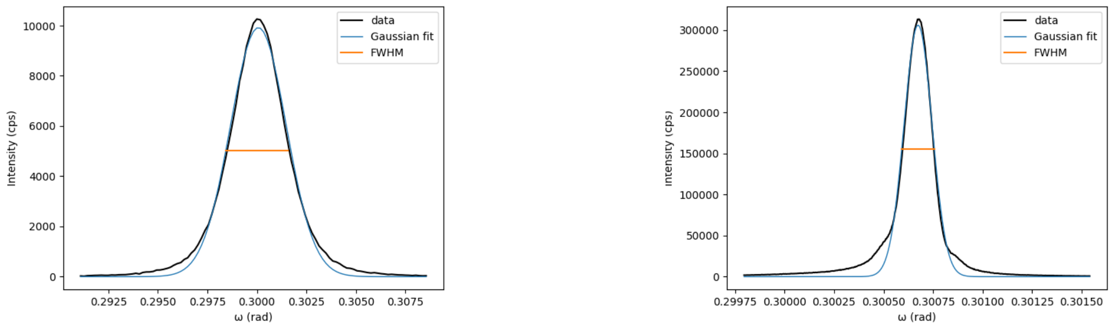

Figure 4: ω scan of sample A and B (002) and Gaussian fit

With $\Delta\omega = 0.0031$ rad for sample A and 0.0002 rad for sample B, lateral sizes $l_{\parallel}$ are 149.114 nm and 2853.987 nm for sample A and B respectively.

By assuming that the broadening of the rocking curves result only from tilt and twist, equation (45) suggests the relation between screw-type dislocation density ρ_{screw} and FWHM $\Delta\omega_{002}$:

$$\rho_{screw} = \frac{\Delta\omega_{002}^2}{4.35|\mathbf{b}_{screw}|^2} \tag{4}$$

with the Burger vector $\mathbf{b}_{screw} = [001]$ which equals to lattice constant $c = 5.2024\mathring{A}$. ρ_{screw} are $8.383 \times 10^{12} m^{-2}$ and $2.286 \times 10^{10} m^{-2}$.

Therefore, sample B has a much larger lateral crystallite size and a much lower screw-type dislocation density than sample A, indicating a better structural quality.

Question 5: *Estimate the edge-type dislocation density of sample A.*

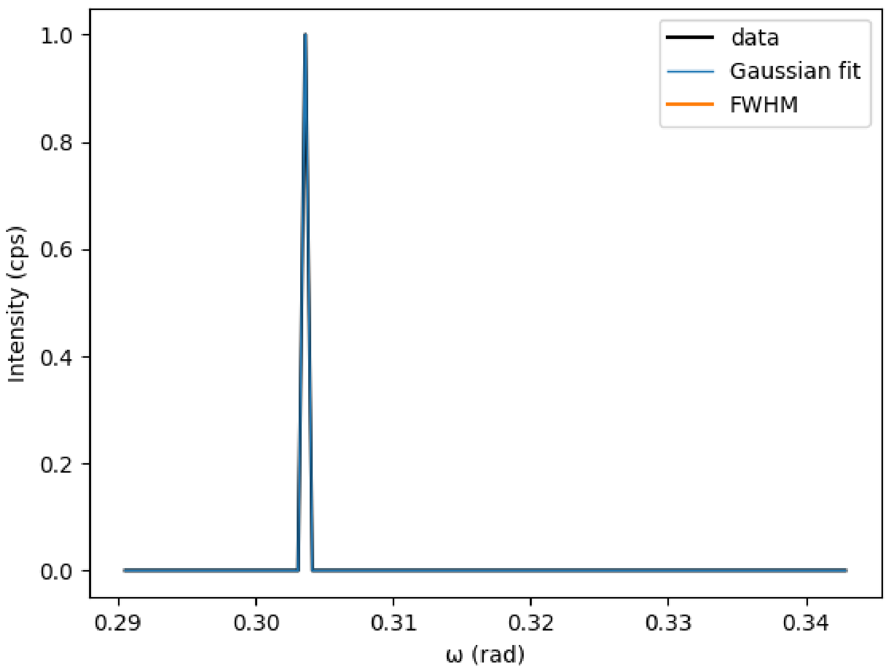

Figure 5: ω scan of sample A (101) and Gaussian fit

By assuming that the broadening of the rocking curves result only from tilt and twist, equation (46) suggests the relation between edge-type dislocation density ρ_{edge} and FWHM $\Delta\omega_{101}$:

$$\rho_{edge} = \frac{\Delta\omega_{101}^2}{4.35|\mathbf{b}_{edge}|^2} \tag{5}$$

with the Burgers vector $\mathbf{b}_{edge} = [110]$ which equals to lattice constance $a = 3.2475\mathring{A}$. However, no broadening is identified in this dataset, which gives 0 FWHM. This suggests that sample A has zero edge-type dislocation density.

Question 6: *Illustrate graphically which lattice planes are associated with the (112) and the (113) reflexes respectively. Verify the epitaxial relation of ZnO on a c-plane sapphire substrate.*

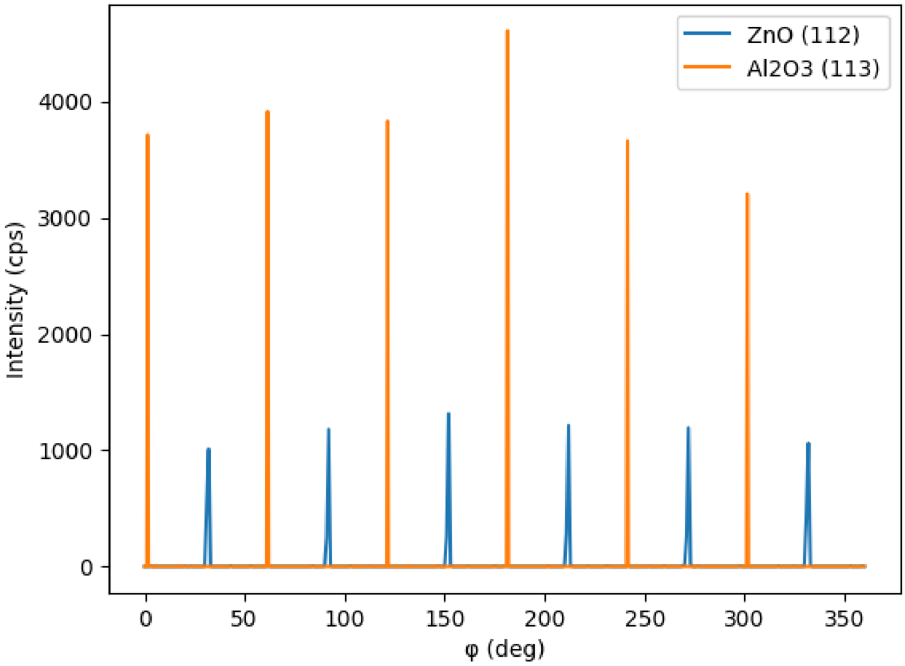

Figure 6: ϕ scan of sample A ZnO (112) and Al_2O_3 (113)

The ϕ scans of the ZnO (112) and Al_2O_3 (113) reflections show six peaks separated by 60°, indicating the hexagonal symmetry of both the film and the substrate. The ZnO peaks are shifted by approximately 30° with respect to the sapphire peaks.

Question 7: *Determine the peak center of the ZnO (205) reflex. Determine the a- and c-lattice constants of sample B. Which mechanisms can you identify for the broadening of its (205) reflex? Calculate the lattice mismatch by assuming that the ZnO film on the sapphire substrate is fully relaxed and by taking into account the formation of a coincidence lattice.*

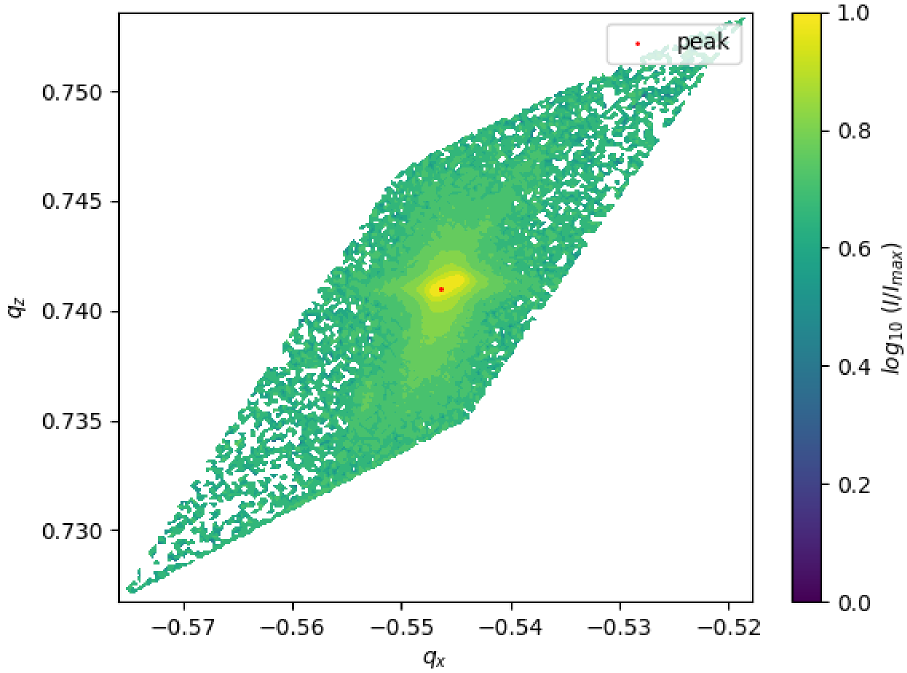

Figure 7: RSM of sample B ZnO (205)

Lattice constants a and c can be calculated by equation (47) and (48):

$$a = \frac{1}{q_{\parallel}(rlu)} \frac{\lambda}{\sqrt{3}} h \tag{6}$$

$$c = \frac{1}{q_{\perp}(rlu)} \frac{\lambda}{2} l \tag{7}$$

where $q_{\parallel}(rlu)$ and $q_{\perp}(rlu)$ are scattering vector q components of the peak center in reciprocal lattice units. Experimentally, a = 0.3255 nm, c = 0.5198 nm.

The broadening results from finite lateral crystallite size and tilt of the crystallites.

The lattice mismatch is defined as:

$$\frac{\Delta a}{a} = \frac{a_{substrate} - a_{film}^{relax}}{a_{substrate}} = \frac{a_{Al_2O_3} - a'_{ZnO}}{a_{Al_2O_3}} \tag{8}$$

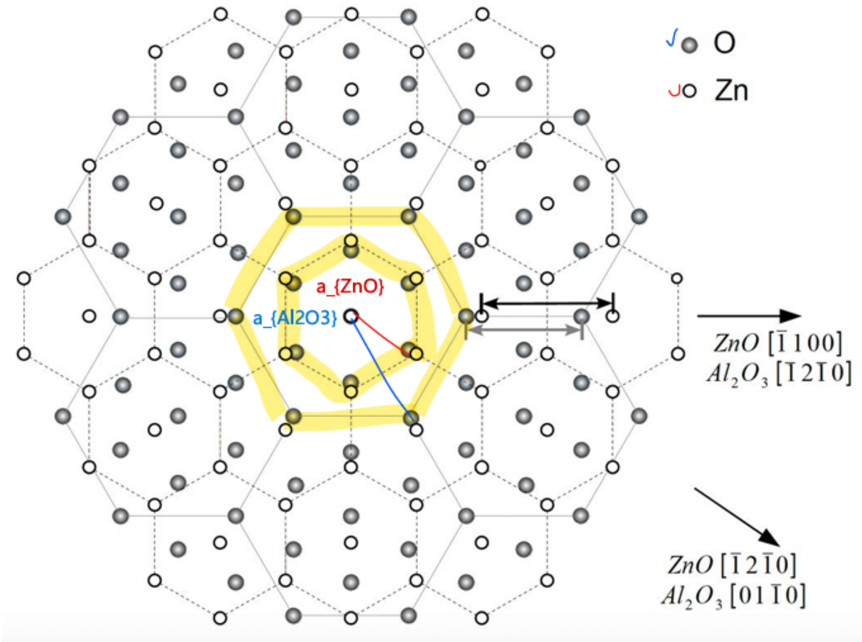

Figure 8: Figure 9 from user guide. Coincidence lattice of ZnO film and Al_2O_3 substrate by 30° rotation.

Considering coincidence lattice, with effective lattice constant along coincidence direction $a'_{ZnO} = 2a_{ZnO}\cos 30° = 0.5639$ nm from experiment, and $a_{Al_2O_3} = 0.47577$ nm from appendix B, lattice mismatch is 18.5%. Relation between two lattice constants a is shown in 8.

Question 8: *Determine the c-lattice constant of sample C. Why was the 006-reflex instead of the (002) reflex used for this purpose? …Use this relation with the above determined value for cZnO to calculate the Mg-content of sample C. Estimate the error of the obtained result by assuming that the Equation (49) is accurate.*

The (006) reflection was used because higher-order reflections appear at larger diffraction angles θ, which improves the accuracy in determining the lattice constant. From equation (39):

$$\frac{\Delta\lambda}{\lambda} = \cot\theta \cdot \theta + \frac{\Delta d_{hkl}}{d_{hkl}} \tag{9}$$

one can see that higher θ suppresses $\Delta\theta$ term.

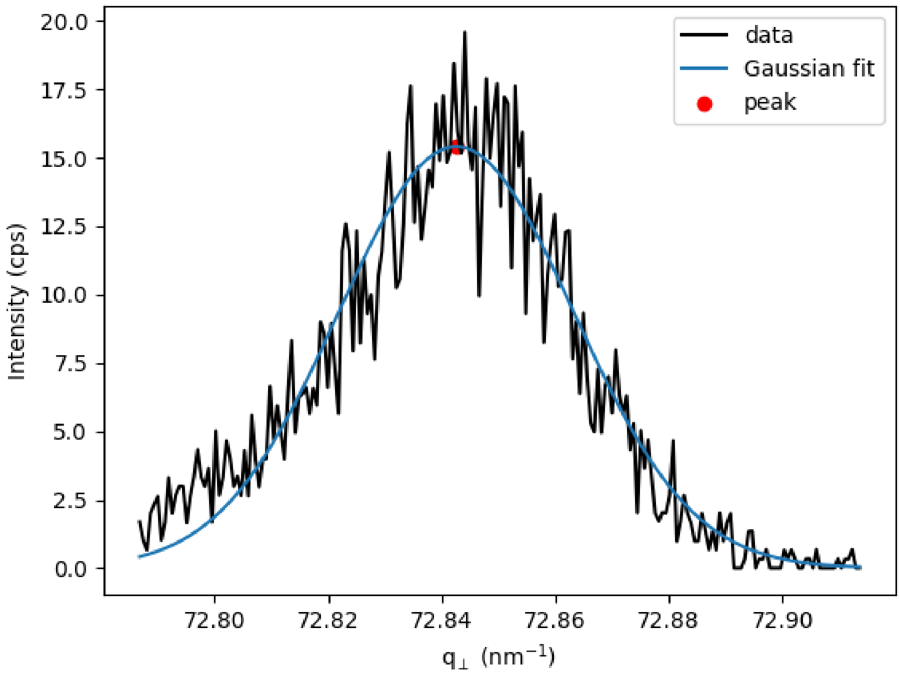

Figure 9: $2\theta/\omega$ scan of sample C (006) as function of $q_\perp$

Calculate c using equation (7), by converting 2θ into $q_\perp$:

$$q_\perp = \frac{2\pi}{\lambda}(\cos(2\theta - \omega) - \cos\omega) \tag{10}$$

The experimental data shows $c = 0.5175$ nm.

Using the relation between c-lattice constant and the Mg-content x for $Zn_{1-x}Mg_xO$ by Sadofev et al. [1]:

$$c_{Zn_{1-x}Mg_xO} = c_{ZnO} - x \cdot 0.17\mathring{A} \tag{11}$$

The experimental data shows $x = 0.1587 \pm 1.6791 \times 10^{-5}$. The error is calculated by error propagation:

$$\delta x = \frac{\partial x}{\partial q_{\perp,peak}} \delta q_{\perp,peak} = \frac{2\pi l}{0.17 q_{\perp,peak}^2} \delta q_{\perp,peak} \tag{12}$$

References

[1] S. Sadofev, S. Blumstengel, J. Cui, J. Puls, S. Rogaschewski, P. Schäfer, Y. G. Sadofyev, and F. Henneberger, *Growth of high-quality ZnMgO epilayers and ZnO/ZnMgO quantum well structures by radical-source molecular-beam epitaxy on sapphire.* Applied Physics Letters, 87:091903, 2005.